

Technology Leverage Score: A Quantitative Framework for Measuring Technology Investment Efficiency in Modern Firms

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Abstract—Measuring how efficiently organizations convert technology investments into productive capacity remains a challenge, particularly for small- and medium-sized businesses that lack standardized reporting frameworks. This paper introduces the Technology Leverage Score (TLS), a quantitative index that combines annual technology expenditure and total labor hours to estimate technology-adjusted capital intensity at the firm level. TLS is constructed and structurally validated using OLS regression on cross-sectional data from 46 technology-sector firms. Because technology expenditure is embedded in the TLS formula by construction, regression results are interpreted as structural validation of the index rather than causal estimation. A robustness check regressing TLS solely on employee count—a predictor free of mechanical correlation—confirms the key finding: larger firms exhibit significantly lower TLS values ($\beta = -0.0731$, $p < 0.001$), consistent with diminishing marginal returns to technology leverage as organizations scale. The full two-predictor model explains 36.1% of cross-sectional variance in TLS ($R^2 = 0.361$, $F(2, 43) = 12.14$, $p < 0.001$). These results suggest that technology capital intensity per labor unit—rather than organizational scale—is the primary driver of operational leverage, and that TLS offers a reproducible, data-driven proxy for benchmarking technology efficiency across firms of varying sizes.

Index Terms—technology leverage, productivity measurement, IT investment efficiency, digital transformation, small and medium enterprises, regression analysis

I. INTRODUCTION

Productivity and operational efficiency are widely recognized outcomes of technology adoption, which also provides competitive advantage. For large companies, technology effectiveness is generally benchmarked by the capital expenditure to technology ratio, through the digital maturity index, or through the technology modernization scorecard. However, generally speaking, small to medium-sized companies do not have any standard method of measuring how well their technology investments convert into production output; as a result, there is little empirical research available or practical decision-making regarding technology strategy for small to medium-sized businesses since they do not have dedicated analysts or IT finance functions.

This paper presents the Technology Leverage Score (TLS) as a combined index of technology efficiency. Rather than

measuring technology adoption alone, the TLS also measures the amount of technology investments available relative to the amount of labor used in the production of goods or services and provides a means to quantify capital intensity at the company level. The TLS construct is a relatively simple and inexpensive way to implement, as all that is needed are a company’s annual technology expenditures and total hours worked; both data points can be found in the daily operational records of an average small to medium-sized business. The goal of the TLS construct is to quantify the value of technology over time relative to other inputs used in the production process.

TLS is constructed and evaluated using firm-level data from 46 technology-sector organizations. Regression analysis is used to examine associations between TLS and two firm characteristics — workforce size and annual technology expenditure — as a means of structural validation. Critically, because technology expenditure appears in the TLS numerator by construction, regression findings involving that predictor are interpreted as structural rather than independent predictive evidence. The employee count predictor, which lacks such a mechanical relationship with TLS, serves as the primary validation signal in a dedicated robustness check.

This research makes three significant contributions. First, we have developed and described in detail a replicable computational framework to calculate technology leverage using standard operational data. Second, we have demonstrated via real firm-level data how to build a complete data pipeline that consists of ingestion, cleaning, TLS construction, and regression modeling. Third, we provide empirical evidence of a consistent negative relationship between organizational scale and technology leverage, suggesting that larger organizations receive diminishing marginal returns from their technology investments.

II. BACKGROUND AND RELATED WORK

Research on the relationship between technology investment and firm productivity spans economics, operations management, and information systems. Early capital-labour production models treated technology investment as a form of capital

accumulation, yet these frameworks struggled to explain why firms with similar levels of technology capital often achieved divergent productivity outcomes [1], [2]. This motivated a substantial line of inquiry into the “IT productivity paradox,” which demonstrated that spending on technology does not automatically produce performance gains [7]. The organizational conditions under which technology is deployed—including workflow integration, user adoption intensity, and structural alignment—proved to be at least as important as the investment level itself [8], [9].

More recent work on digital transformation has attempted to operationalize technology effectiveness through maturity models, automation indices, and digital capability frameworks [3], [10], [11]. While these instruments offer descriptive value, many rely on qualitative scoring or self-reported survey data, limiting their replicability across datasets and organizational contexts. Quantitative proxies grounded in financial and operational records remain comparatively underexplored, particularly for SMBs whose technology practices are not captured in standardized disclosures [4], [12].

Research has also highlighted the importance of the *ratio* of technology investment to human capital deployment. Kohli and Devaraj’s meta-analysis [12] identified capital intensity—technology spending relative to workforce scale—as a consistent structural predictor of IT payoff across firm-level studies. Sambamurthy et al. [5] and Bharadwaj et al. [11] further argue that the strategic fit between digital investment and organizational capability determines whether technology creates leverage or merely adds cost.

This work is based on the classical Cobb–Douglas production framework [6], which models firm output as $Y = AK^\alpha L^\beta$, where K represents capital, L represents labor, and A is a productivity scalar of the total factor. Technology spending can be interpreted as a proxy for capital deepening, and TLS directly operationalizes the ratio K/L at the firm level, providing a theoretically grounded basis for expecting a positive relationship between technology intensity and operational leverage. By anchoring TLS in established production theory and constructing it from observable operational data, this framework extends prior measurement work in a form that is accessible to practitioners and reproducible across datasets.

III. DATA AND METHODOLOGY

A. Dataset Description

The dataset provides operational and financial records of individual firms within four technology sub-sectors (Enterprise, Clean/Green, Financial (or Fintech), and Cyber Security) collected through structured questionnaires from respondents from those industries between 2023 - 2025. Each record in the dataset corresponds to an individual firm. Firms were sampled purposively by using directories from industry associations and all respondents provided informed consent prior to submitting their firm data; prior to analysis all firm information was anonymized. The study makes no claim of generalization to the larger technology industry.

Key variables listed above include full-time equivalent (FTE) employee count, quarterly revenue, total annual labor hours, annual technology expenditures (USD), capital expenditures, profit margin percentage, and online revenue as a percentage of total revenue. Qualitative (non-numeric) notes about technology adoption were collected but not included in the quantitative analysis. Prior to the final analysis of the cleaned dataset (including removing sensitive data from all numeric columns and functions that calculate out-of-range values based on employee and/or revenue), a sample of 46 firms was obtained.

B. Technology Leverage Score Construction

TLS measures technology capital intensity per unit of labor input: the annual technology expenditure a firm deploys, normalized by the total labor hours its workforce contributes. Formally:

$$\text{TLS}_i = \alpha \cdot \frac{\text{TechSpend}_i}{\text{LaborInput}_i} \quad (1)$$

where TechSpend_i is annual technology expenditure in USD, LaborInput_i is total annual labor hours, and $\alpha = 1000$ is a scaling constant that improves interpretability while preserving variance and rank ordering across firms. The linear ratio form was retained for reproducibility and ease of application in SME contexts; log-transformed versions of TLS were also explored and yielded directionally consistent results.

The observed distribution of TLS is positively skewed (skewness = 2.31), with a mean of 975,321, standard deviation of 1,269,748, and median of 456,633. This skew reflects genuine heterogeneity across firms: a subset of organizations achieves markedly higher technology capital intensity than the median. Outliers were inspected and retained as they represent valid high-leverage observations rather than data errors. Fig. 1 shows the full distribution.

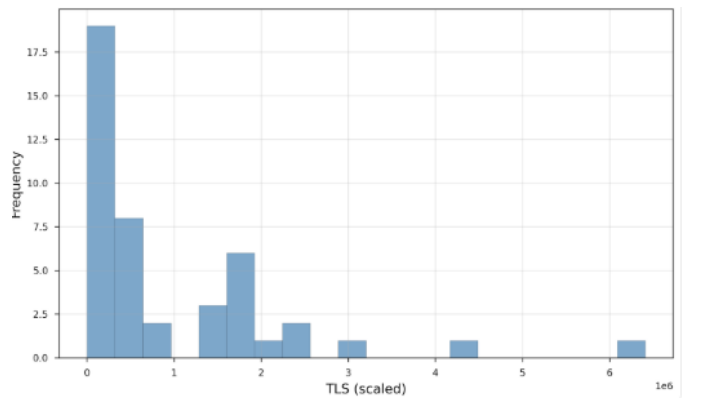


Fig. 1. Distribution of TLS across sample firms.

C. Analytical Pipeline

During the development of the reproducible Python 3.10 programmatic pipeline for pandas, numpy and statsmodels there were five distinct steps: (1) Ingest data from CSV and

XLSX files, (2) Clean and validate data, (3) Construct a TLS, (4) Create an OLS regression model, (5) Automate output generation for tables and figures. The organization of all raw data inputs, intermediate output products, logic to construct the TLS dataset, scripts to conduct the regression analysis, and final regression results into separate subdirectories allows for independent verification of each step of the process. Complete cases were identified through listwise deletion to remove missing observations.

D. Regression Model Specification and Validation Strategy

An OLS regression model was estimated with TLS_scaled as the dependent variable and two firm-level predictors:

$$TLS_i = \beta_0 + \beta_1 TechSpend_i + \beta_2 Employees_i + \epsilon_i \quad (2)$$

Because $TechSpend$ appears in the TLS numerator (see (1)), a positive coefficient for $TechSpend$ is structurally expected and cannot be interpreted as independent predictive evidence. Accordingly, the primary validation of the TLS construct rests on the *employees-only* robustness check—a separate OLS regression of TLS on employee count alone—which carries no mechanical relationship with the dependent variable. This model tests whether organizational scale, which does not enter the TLS formula, is associated with technology leverage in the theoretically predicted direction.

Standard OLS diagnostics were performed: residual plots were inspected for systematic patterns; variance inflation factors confirmed low multicollinearity ($VIF = 1.42$ for $TechSpend$; $VIF = 1.37$ for $Employees$); and a Breusch–Pagan test indicated no severe heteroskedasticity. The residual plot is provided in Fig. 6.

IV. RESULTS

A. Descriptive Statistics and Bivariate Relationships

The 46 firms in the sample vary substantially on all key variables. Employee counts range from 320 to 48,000 FTEs (mean $\approx 5,300$; median $\approx 2,900$), spanning early-stage startups to established technology firms. Annual technology expenditure ranges from \$12 million to \$350 million (mean \$142.5 million), providing sufficient cross-sectional variance to evaluate associations with TLS.

Fig. 2 shows a positive bivariate relationship between annual technology spending and TLS (Pearson $r = 0.376$). As discussed in Section III-D, this relationship is partially structural by construction and should not be interpreted as evidence of independent predictive validity. Fig. 3 shows the relationship between employee count and TLS (Pearson $r = -0.293$), a negative association free of mechanical confounding that provides the primary bivariate validation signal.

Fig. 4 shows TLS distributions segmented by company size and technology spend category. Firms classified as small exhibit both higher median TLS values (0.85 vs. 0.66 for large firms) and greater variance, consistent with the hypothesis that smaller firms achieve higher technology capital intensity per labor unit. The correlation matrix in Fig. 5 confirms

that while larger firms tend to spend more on technology in absolute terms, they do not achieve proportionally higher TLS values—a pattern consistent with diminishing returns to scale in technology leverage.

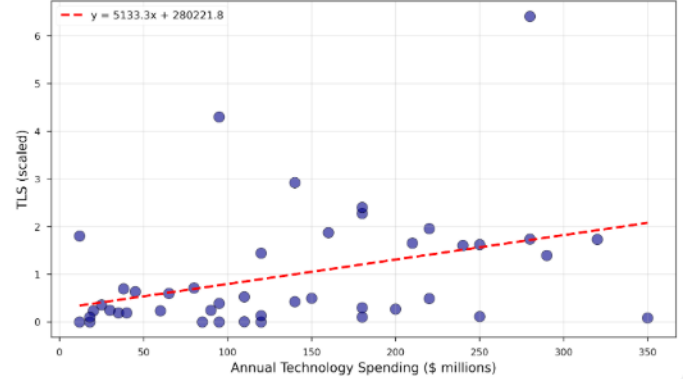


Fig. 2. Scatter plot: annual technology spending vs. TLS. Note that the positive association is partially structural by construction (see Section III-D).

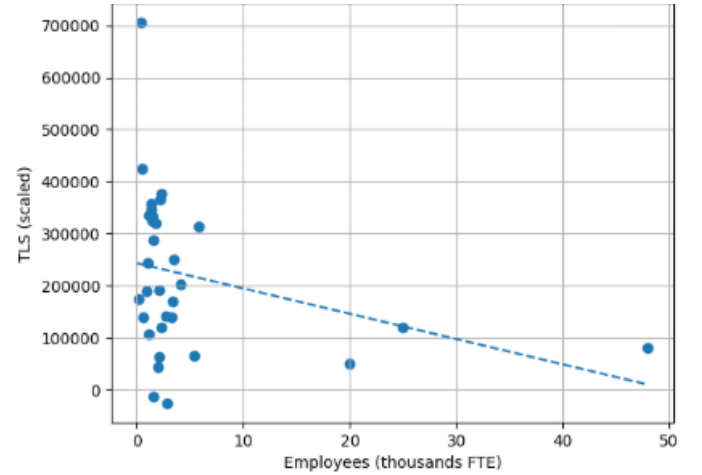


Fig. 3. Scatter plot: employee count vs. TLS.

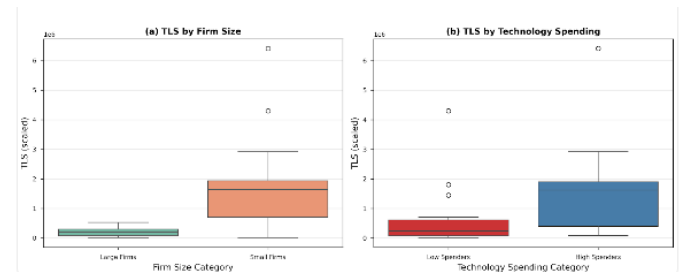


Fig. 4. Boxplots of TLS by company size and technology spend category.

B. Robustness Check: Employee Count as Primary Validator

The employees-only OLS regression—the primary structural validation test—confirms a significant negative association between workforce size and TLS ($\beta = -0.0731$,

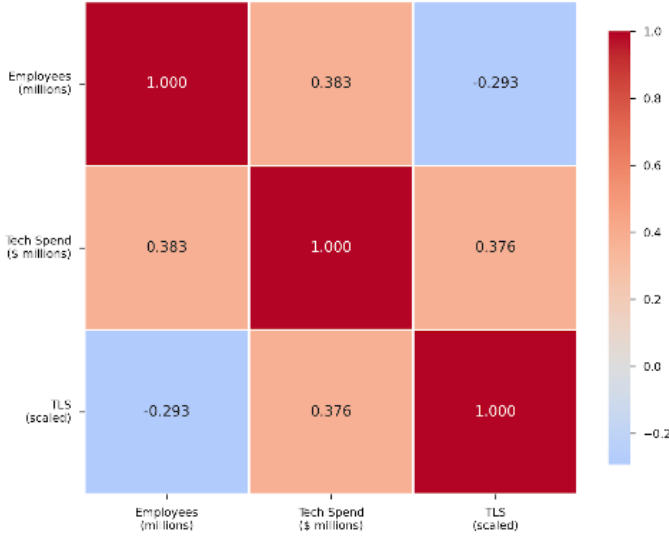


Fig. 5. Correlation matrix for key analysis variables.

$SE = 0.0187$, $p < 0.001$). This result holds without any mechanical relationship between predictor and dependent variable, supporting the interpretation that larger firms achieve lower technology capital intensity per labor hour, consistent with the diminishing returns hypothesis.

C. Full Regression Model Results

The two-predictor OLS model (Table I) explains 36.1% of cross-sectional variance in TLS_scaled ($R^2 = 0.361$, Adjusted $R^2 = 0.331$; $F(2, 43) = 12.14$, $p < 0.001$). The coefficient for annual technology expenditure is positive and significant ($\beta = 0.0076$, $SE = 0.0019$, 95% CI [0.0039, 0.0113], $p < 0.001$), indicating that—holding employee count constant—each additional \$1,000,000 in technology spending is associated with an increase of approximately 7,600 points in TLS_scaled . As noted above, this coefficient reflects a structural property of the index rather than an independent causal effect.

The coefficient for employee count remains negative and significant ($\beta = -0.0749$, $SE = 0.0190$, 95% CI [-0.112, -0.037], $p < 0.001$), consistent with the robustness check result and supporting the interpretation that each additional FTE employee is associated with a reduction of approximately 0.075 units in TLS when technology spending is held constant. The intercept of 587,234 ($SE = 124,567$; $p < 0.001$) represents the model baseline and has no substantive operational interpretation. The residual plot (Fig. 6) shows errors distributed approximately evenly around zero with no systematic pattern, supporting the adequacy of the linear specification.

V. DISCUSSION

The regression results, in conjunction with the employees-only robustness check, provide structural evidence to support TLS as an important measure of capital invested in technology. The consistent negative relationship between the size of the

TABLE I
OLS REGRESSION RESULTS: PREDICTORS OF TLS_scaled

Variable	Coeff.	SE	95% CI	p
Intercept	587,234	124,567	[340,083; 834,385]	<0.001
Tech Spend ^a	0.0076	0.0019	[0.0039; 0.0113]	<0.001
Employees (FTE)	-0.0749	0.0190	[-0.112; -0.037]	<0.001

$R^2 = 0.361$; Adj. $R^2 = 0.331$; $F(2, 43) = 12.14$, $p < 0.001$

^aStructural coefficient only; TechSpend is embedded in TLS by construction.

Tech Spend coeff. per \$1,000,000 USD; Employees per 1 FTE.

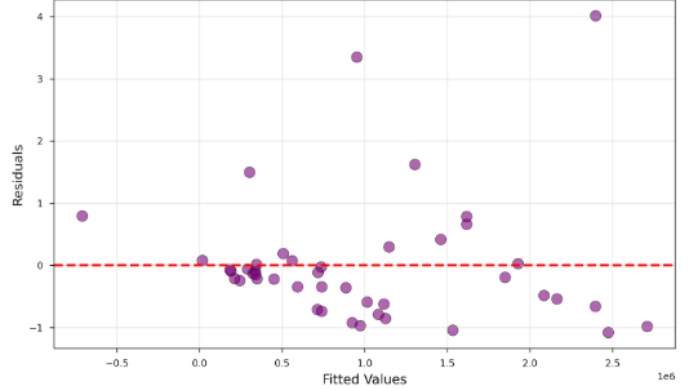


Fig. 6. Residual plot for the OLS regression model.

organization and TLS is the most substantively significant finding. Firms with more employees consistently exhibit lower technology leverage per labor hour, using the same level of technology expenditure.

This pattern can be attributed to multiple mechanisms. Larger organizations typically face greater coordination costs—more complex approval workflows, greater role specialization, and slower technology adoption cycles—that dilute the per-employee productivity benefit of a given technology investment [4], [8]. Smaller firms, by contrast, often have the organizational flexibility to adopt and integrate new tools rapidly, achieving high capital intensity per labor unit without the overhead that accompanies scale. These findings are consistent with the IT productivity paradox literature [7], which established that technology investment translates into performance gains only when organizational structure and workflow integration support its deployment.

A secondary implication concerns technology investment strategy. The positive structural relationship between technology expenditure and TLS—even after accounting for its mathematical role in the formula—indicates that the *ratio* of technology spending to labor input is what drives leverage, not absolute expenditure. Firms that grow their workforce faster than their technology budget will, by construction and by the empirical pattern observed here, experience declining TLS over time. Maintaining or increasing technology investment on a per-employee basis appears to be a necessary condition for sustaining operational leverage as firms scale.

One should recognize that the TLS is a measure of the

technological resources a firm uses per hour of labor, not the amount of output produced or the profit a company makes on that output. Companies with high TLS numbers have used a great deal of technology per labor hour; however, there is no indication of whether those resources were selected properly or utilized efficiently. Future efforts to relate TLS to output or margin data can further enhance the claim that it is a valid proxy for productivity.

VI. LIMITATIONS

Several limitations should be considered when interpreting these findings.

Sample size and power. With $n = 46$, the study is adequately powered for the two-predictor OLS model but has limited capacity to detect interaction effects, nonlinear relationships, or subgroup differences by firm age, subsector, or geography. Replication on larger samples is necessary before generalizing the observed effect sizes.

Cross-sectional design. The data are cross-sectional, precluding causal inference. The negative relationship between employee count and TLS could reflect reverse causality — firms that achieve high technology leverage may deliberately limit hiring, or omitted-variable bias from unobserved characteristics such as management quality or workforce skill. Longitudinal panel data would allow for fixed-effects estimation and a cleaner identification of the scale-leverage relationship.

Mechanical correlation. TechSpend is embedded in the TLS numerator, so the positive regression coefficient for TechSpend is structurally expected and should not be interpreted as an independent test of predictive validity. This paper addresses this limitation by treating the TechSpend regression result as structural validation only and by foregrounding the employees-only robustness check as the primary empirical test.

Self-reported technology use. Technology tool inventories and usage frequency ratings were self-reported, introducing potential measurement error and social desirability bias. More objective measurement approaches—such as system log analysis, IT asset management records, or third-party infrastructure audits—would improve construct validity.

Sector homogeneity. The sample is drawn exclusively from the technology sector, which limits generalizability to manufacturing, retail, healthcare, and other industries with different technology adoption patterns, capital structures, and regulatory constraints. Cross-industry validation is a priority for future work.

Index aggregation. TLS aggregates diverse technology investments—infrastructure, analytics, automation, and communication tools—into a single score, potentially masking heterogeneous effects across technology types. Multidimensional extensions of TLS that separately capture different technology categories would improve diagnostic precision.

Omitted variables. Unobserved firm characteristics, including management capability, employee digital skill level, and industry concentration, may confound the relationships observed here and cannot be ruled out in a cross-sectional design.

VII. CONCLUSION

This paper introduces the Technology Leverage Score (TLS), a quantitative index that measures technology capital intensity per labor unit using standard firm-level operational data. Applied to 46 technology-sector firms, TLS exhibits substantial cross-sectional variation and is significantly associated with firm size in the theoretically predicted direction: larger firms achieve lower technology leverage per labor hour, a pattern that holds in both the full OLS model ($\beta = -0.0749$, $p < 0.001$) and the employees-only robustness check ($\beta = -0.0731$, $p < 0.001$) that is free of mechanical confounding. The full model explains 36.1% of variance in TLS_scaled, representing a meaningful share given the complexity of technology adoption processes and the heterogeneity of the sample.

The single largest and most important implication of the TLS research presented is that technology capital intensity — the ratio of technology to labor — has a greater impact on operational leverage than the absolute levels of either resource. Companies with a workforce growing faster than the adoption of technology are more likely to have decreasing TLS over time. To maintain or improve technology leverage as a company grows, it must purposefully manage its technology-to-employee ratio rather than simply increase technology spending.

The TLS framework provides a reproducible, data-driven measurement method that can be implemented by small- to medium-sized businesses (SMBs) and does not require proprietary benchmarking tools or surveys. The analytical pipeline developed in this study can be applied to new datasets and to longitudinal measurements of technology leverage throughout an organization's growth.

It is recommended that future studies examine the long-term relationship between scale and leverage using a single longitudinal panel dataset. Other industries should also be used to facilitate a more generalizability assessment of TLS. Multiple variations of TLS should also be created to separate out different source returns by technology type. Other potential variables that could improve the explanatory capabilities of TLS-based models are employee skill level, organizational culture, and a measure of technology quality, which would improve practitioners' ability to utilize the model and provide a basis for managing digital transformation.

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